

Jack Kolb

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Education	Georgia Institute of Technology , Atlanta, GA	
	<i>PhD in Robotics</i>	2025
	<i>MS in Computer Science</i>	2023
	Advisor: Karen Feigh Committee: Julie Adams, Sonia Chernova, Harish Ravichandar, Alan Wagner	
	University of California at Riverside , Riverside, CA	
	<i>BS in Mechanical Engineering (Cum Laude, Honors)</i>	2020
Work Experience	Apple	June 2025 - present
	<i>Senior Research Engineer/Scientist, Special Projects</i>	
	• Under NDA Leading research and development projects for a new device category. • <i>Directly Responsible Individual</i> for two major system components, owning their algorithm design, crossfunctional collaboration, implementation, and evaluation metrics. • Initiated and led a substantial system re-architecture that addressed longstanding issues, enabled novel device capabilities, and enhanced user interaction fluency. • Interviewed candidates for various robotics roles, evaluating perception algorithms, semantic reasoning, task & behavior planning, modeling people, and system design.	
	Travelers	Mar 2025 - June 2025
	<i>Data Scientist (Intern), Emerging AI & Technologies</i>	
	• Worked on verification and alignment for the (now) award-winning TravelersLLM, a custom model trained on millions of insurance documents to support underwriters. • Created an LLM harness to identify claims in responses, locate supporting documentation from internal company data lakes, and flag potentially incorrect claims.	
	Gatik AI	May 2022 - Aug 2022
	<i>Robotics Engineer (Intern), Behavior Prediction</i>	
	• Designed and prototyped a novel graph neural network architecture for forecasting surrounding vehicle trajectories, tailored to Gatik’s operational design domain. • Presented design recommendations for Gatik’s surrounding vehicle prediction module in a company-wide talk.	
	NextGen Assistive Technologies	May 2020 - Aug 2020
	<i>Software Engineer (Intern)</i>	
	• Developed the minimum viable product for a sensor-based smart home remote caregiving system, now deployed in 100+ homes and producing \$500k+ ARR. • Integrated sensor event intake, video conferencing, and event resolution, enabling each caregiver to support 10+ clients via remote monitoring and interactions.	
	UC Riverside Marine Robotics (RoboSub)	Sept 2016 - Aug 2020
	<i>Project Lead ('18-20)</i>	
	<i>Mechanical Team Lead ('17-18)</i>	
	<i>Mechanical Team Member ('16-17)</i>	
	• Competed in the international RoboSub competition (2018, 2019, 2020). • Led 25+ members in the development of two autonomous marine robot platforms for aquatic navigation and interaction tasks. • Designed and implemented software systems for underwater autonomy, navigation, vision, mission control, and object interaction. • Designed and manufactured physical hardware, wrote field test plans and procedures, and managed relationships with sponsors.	

Research Experience	Cognitive Engineering Center <i>Graduate Research Assistant</i> Georgia Tech Sept 2022 - May 2025 • Investigated how household robots can infer and apply a person’s world belief state for active assistance, team reasoning, and informed communication. • Applied LLMs and classical methods to reason over estimated user belief states and summarize state information in task-driven collaborative human-AI teams. • Researched generative image manipulation of poses using text-guided instructions. • Developed autonomous aircraft wingmen using deep reinforcement learning and constrained control to support human pilots in collaborative high-risk missions. • Explored structuring shared human-AI decision-making processes to enhance user situational awareness and decision outcomes. • Wrote and awarded an \$80k grant from Amazon Consumer Robotics (Lab126). • Wrote papers accepted to leading robotics venues – RA-L, IROS, RO-MAN.
	Robot Autonomy and Interactive Learning Lab <i>Graduate Research Assistant</i> Georgia Tech Sept 2020 - May 2022 • Applied machine learning to predict user teleoperation task performance and inform role assignment, improving team performance by 24%. • Improved the sample efficiency of dexterous robot “pick and place” manipulation by structuring human demonstrations and offline training. • Wrote quarterly reports and presented research talks to grant sponsors. • Wrote papers accepted to leading robotics conferences – IROS, RO-MAN.
	Sundararajan Venkatadriagaram Research Group <i>Undergraduate Research Assistant</i> UC Riverside Mar 2018 - June 2020 • Designed and prototyped an IOT sensor network to record and analyze vibrations of electric motors to classify motor damage and predict mechanical failure. • Tried system on university campus ventilation and water infrastructure, worked with university to commercialize system.
Tools Languages	Git, LangChain/SGLang, Linux, OpenCV, PyTorch, ROS, scikit-learn, Unity3D Python, C++, C#, Go, HTML/JS, Swift
Awards	Amazon Consumer Robotics Grant (\$80k), 2022 <i>Awarded to support my dissertation research on enabling robots to estimate a user’s belief state in household human-robot teaming domains.</i>
	IEEE RO-MAN – Best Student Paper Finalist (3/237, 1.2%), 2022 <i>For my work on applying user cognitive skills to predict user performance at robot teleoperation tasks and inform role assignment.</i>
	HackGT – Winner, 2021 <i>Awarded for RoboVR: a multi-user platform for teleoperating 10+ real-world robots in virtual reality.</i>
	UC Riverside – Best Large-Scale Student Project, 2019 <i>Accepted on behalf of UCR Marine Robotics, for our work in designing and developing autonomous underwater vehicles.</i>
	CitrusHack – Winner, 2018 <i>Awarded for BlindSight: a hat that enabled “feeling” the proximity of surrounding objects through localized and directioned haptic vibrations.</i>
Professional Activities	Teaching Assistant • CS6476: Computer Vision (Graduate Level) Sp’24, Fa’24, Sp’25 • CS6262: Network Security (Graduate Level) Su’24

Reviewing

- HFES ASPIRE (conference) 2023-2026
- IEEE BioRob (conference) 2024
- IEEE/ACM HRI (conference) 2024, 2026
- IEEE ICRA (conference) 2025
- IEEE RO-MAN (conference) 2024
- RSS (conference) 2025

Invited Talks

- Georgia Tech Oct 2025
Bridging the Gap Between Tools and Teammates for Robots in Human-Centered Environments
- Georgia Tech Feb 2025
Inferring World Belief States in Human-Robot Teaming
- Exponent, Inc. Feb 2025
Unraveling the Black Box: How systems can leverage cognitive, physiological, and mental states to improve human-AI teaming.

Advising

Formal mentoring of students on research projects.

- Ryan Bowers (MS at GaTech → PhD at GaTech) 2024 - 2025
Using deep reinforcement learning for controlling autonomous wingmen.
- Alagappan Swaminathan (MS at GaTech → PhD at GaTech) 2023 - 2025
User belief state estimation for human-swarm command & control.
- Richard Agbeyibor (PhD at GaTech) 2022 - 2025
Adaptive autonomy for human-AI systems and autonomous wingmen.
- Sanya Doda (PhD at GaTech) 2022 - 2025
Real-time cognitive workload assessment from biometric sensors.
- Rohan Shrivastava (BS at Duke) 2024
Identifying misinformation at internet-scale for computational anthropology.
- Pranav Gopalabhatla (BS at Purdue) 2023
Predicting asthma prevalence from air quality and environmental factors.
- Sia Godika (BS at MIT) 2023
Predicting malaria incidence in underdeveloped regions.
- Mayank Kishore (MS at GaTech → Founder at Mirage ML) 2021 - 2022
Virtual human-robot command & control tasks.

Journal Papers

2. **Inferring Teammate Belief States in Human-Robot Teams**
J. Kolb, A. Garg, N. Warner, K. Feigh.
Under review.
1. **Attention-Tunable Safety Barriers for Fluent Human-Autonomy Coordination in Specialized Aviation Missions**
C. Cortes, R. Agbeyibor, J. Kolb, K. Feigh, S. Coogan.
IEEE Robotics and Automation Letters (RA-L), 2025.

Conference Proceedings (selected)

8. **Model Cards for AI Teammates: Comparing Human-AI Team Familiarization Methods for High-Stakes Environments**
R. Bowers, R. Agbeyibor, J. Kolb, K. Feigh.
IEEE Intl. Conf. on Robot & Human Interactive Communication (RO-MAN), 2025.
7. **Learning Complex Non-Rigid Image Edits from Multimodal Conditioning**
N. Warner, J. Kolb, M. Hahn, J. Huang, I. Essa, V. Birodkar.
IEEE International Conference on Image Processing (ICIP), 2025

6. **Inferring Belief States in Partially-Observable Human-Robot Teams**
J. Kolb, K. Feigh.
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024.
5. **Human-AI Collaboration in Autonomous Aerial Vehicles for ISR: Experience, Trust, and Perception**
R. Agbeyibor, V. Ruia, J. Kolb, K. Feigh.
HFES International Annual Meeting (ASPIRE), 2024.
4. **Towards Safe Collaboration Between Autonomous Pilots and Human Crews for Intelligence, Surveillance, and Reconnaissance**
R. Agbeyibor, V. Ruia, J. Kolb, C. Cortes, T. Mancao, S. Coogan, K. Feigh.
IEEE/AIAA Digital Avionics Systems Conference (DASC), 2024.
3. **The Effects of Robot Motion on Comfort Dynamics of Novice Users in Close-Proximity Human-Robot Interactions**
P. Howell, J. Kolb*, Y. Liu*, H. Ravichandar.
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023.
2. **The Effects of Inaccurate Decision-Support Systems on Structured Shared Decision-Making for Human-Robot Teams**
J. Kolb, D. Srivastava, K. Feigh.
IEEE Intl. Conf. on Robot & Human Interactive Communication (RO-MAN), 2023.
1. **Leveraging Cognitive States in Human-Robot Teaming**
J. Kolb, H. Ravichandar, S. Chernova. [Best Student Paper Finalist]
IEEE Intl. Conf. on Robot & Human Interactive Communication (RO-MAN), 2022.

**Symposium
& Workshop
Proceedings
(selected)**

6. **Investigating Human-AI Team Fluency in Autonomous Medical Evacuation: A Study of Novice Aviator Cognitive States and HAI Interface Design**
S. Doda, R. Agbeyibor, C. Cortes, J. Kolb, J. Magalhaes, K. Feigh.
AIAA Aviation Forum and Exposition, 2025.
5. **Use of Simulated Mental Models and Active Replanning for Human-Robot Interaction**
J. Ren*, A. Swaminathan*, J. Kolb, Y. Zhao, S. Coogan, K. Feigh.
AIAA SciTech Forum and Exposition, 2025.
4. **A Framework for Inferring Belief States in Partially-Observable Human-Robot Teams**
J. Kolb, K. Feigh.
40th Anniversary of the IEEE Conf. on Robotics and Automation (ICRA@40), 2024.
3. **Run Time Assurance and Human AI Fluency in Manned Autonomous Intelligence Surveillance and Reconnaissance**
R. Agbeyibor, V. Ruia, C. Cortes, J. Kolb, S. Coogan, K. Feigh.
AIAA Aviation Forum and Exposition, 2024.
2. **Safe Dexterous Manipulation Using Geometric Boundary Constraints**
A. Jain*, J. Kolb*, H. Ravichandar.
Safe Reinforcement Learning Workshop at the International Joint Conference on Artificial Intelligence (IJCAI), 2022.
1. **Evaluating the Effectiveness of Corrective Demonstrations and a Low-Cost Sensor for Dexterous Manipulation**
A. Jain*, J. Kolb*, J. Abbess, H. Ravichandar.
Machine Learning in Human-Robot Collaboration Workshop at the International Conference on Human-Robot Interaction (HRI), 2022.